

- 5 (a) Binding energy per nucleon is a maximum at around $A = 56$. Products splitting a Fe-56 nucleus must have a lower total binding energy. (Reaction would require) a net input of energy.

- (b) Total energy released (and momentum) per decay is constant. If there is no other particle emitted, ALL the beta particles should have the same energy. There is a range of energies of β^- particles as shown from the figure. Hence, there must exist another particle, neutrino, to possess/share the remaining energy (and momentum).

(c) (i) Energy released = loss in mass $\times c^2$
 $= (211.9459 - 207.9374 - 4.0015)(1.66 \times 10^{-27})(3.0 \times 10^8)^2$
 $= 1.0458 \times 10^{-12}$
 $= 1.05 \times 10^{-12} \text{ J}$

(ii) KE of α -particle = $(0.25 \times 212 - 1)(\text{KE of Thallium})$
 KE of Thallium = $(1/52)(\text{KE of } \alpha\text{-particle})$

KE of α -particle + KE of Thallium = 1.0458×10^{-12}
 KE of α -particle + $(1/52)(\text{KE of } \alpha\text{-particle}) = 1.0458 \times 10^{-12}$

KE of α -particle = $1.026 \times 10^{-12} \text{ J}$
 $= (1.026 \times 10^{-12}) / (1.6 \times 10^{-19} \times 10^6)$
 $= 6.41 \text{ MeV}$

- (iii) A γ -ray photon is emitted. The emitted γ -ray photon has a certain momentum associated with its wavelength / frequency / energy. Hence, by conservation of momentum, the α -particle and thallium nucleus may not be in exactly opposite directions due to the emission of γ photon.

- (iv) 1. Thallium-208 has a much shorter half-life than Bismuth-212 / much higher decay constant

2. $N_0 = 10 / (209 \text{ u}) = 2.882 \times 10^{25}$ (using u)
 OR $N_0 = 2.88038 \times 10^{25}$ (using N_A)

$A = \lambda N = (\ln 2 / 1.9 \times 10^{19}) 2.882 \times 10^{25} = 1.051 \times 10^6 \text{ yr}^{-1}$

Since 100 years is much smaller than half-life, activity is constant over 100 years. No. of nuclei disintegrated = $A t = 1.051 \times 10^6 \times 100 = 1.05 \times 10^8$

6 (a) Scattering forces tend to push the particle in the direction of the beam. The gradient